Case Study: The Change Healthcare Ransomware Attack

Type: Ransomware

Overview

In early 2024, **Change Healthcare**, a critical subsidiary of UnitedHealth Group that processes nearly 40% of all U.S. medical claims, fell victim to a devastating ransomware attack by the **BlackCat (ALPHV)** group. This incident is considered one of the most significant cyberattacks in the history of the American healthcare system.

How the Crime Happened

The attack followed a sophisticated, multi-stage process:

1. **Initial Access:** Hackers gained entry by exploiting compromised credentials on a remote access application that lacked **multi-factor authentication (MFA)**.
2. **Lateral Movement:** Once inside, the attackers spent nine days moving through the network, identifying high-value databases and exfiltrating **6 terabytes** of sensitive data.
3. **Execution:** On February 21, the group deployed ransomware that encrypted the company's systems, effectively "locking" the digital infrastructure.
4. **Extortion:** A ransom note demanded payment in cryptocurrency. To protect patient data, UnitedHealth reportedly paid a **$22 million ransom**, though a second group later attempted to extort them again using the same stolen data.

Targets and Consequences

The primary target was a massive "clearinghouse," but the secondary victims were **900,000 physicians, 33,000 pharmacies, and 5,500 hospitals**.

The consequences were catastrophic:

- **Operational Paralysis:** Doctors could not verify insurance or submit claims, forcing many to use pen and paper or dip into personal savings to stay afloat.
- **Financial Impact:** The breach cost UnitedHealth over **$2 billion** in recovery and lost revenue.
- **Privacy Breach:** Sensitive medical and personal data of an estimated **190 million individuals** were compromised, leading to a massive risk of long-term identity theft.

Source: [HHS Office for Civil Rights / UnitedHealth Group 2024-2025 Investigative Reports]